PseudoCode

```
Function missingInteger_iterative(A[0 .. N-1])

  Size = N+2

  exist [0 .. size-1] = all 0

  For i = 0 to N-1 :
  IF A[i] > 0 And A[i] < Size :
   exist [A[i]] = 1
   End IF
 End For

  For i=1 To Size - 1 :
    IF exist[i] = 0 :
     Return i
      End if
  End for
```

Analysis

* Basic operation is the Comparison

if (A[i] > 0 && A[i] < size)

Best Case:

Loop 1: N+2
Loop 2: N
★ Loop 3: 1

Total = 2N + 3 ⇒ O(n)

Worst Case:

Loop 1: N+2
Loop 2: N
★ Loop 3: N+1

Total = 3N + 3 ⇒ O(n)

مش N+2
لأن الـloop بتبدأ من
i=1 مش من i=0

The Best case occures when only number 1 isn't in the array.

The Worst case occures when the array contains exactly 1, 2, 3, ..., N.